A Contrastive Learning - AU Level F1 Scores

Table 6: Comparison of AU Level Recognition Performance

AU	Without Contrastive Loss	With Contrastive Loss
AU1	52.50	**64.85**
AU2	45.70	**58.32**
AU4	**76.10**	72.15
AU6	51.80	**52.71**
AU9	46.50	**54.00**
AU12	**76.10**	69.44
AU25	92.90	**95.25**
AU26	**57.60**	53.70
Average	**62.40**	**65.05**

Table 7: Comparison of OELE AU Recognition Performance With and Without Contrastive Learning

AU	Without Contrastive Loss	With Contrastive Loss
AU1	44.51	**45.66**
AU2	65.46	**67.17**
AU4	73.77	**79.32**
AU5	42.89	**53.37**
AU6	69.41	**70.10**
AU7	45.87	**50.35**
AU9	44.13	**51.09**
AU14	28.69	**30.41**
AU15	17.04	**28.72**
AU17	**41.15**	32.60
AU18	53.01	**54.20**
AU24	**47.81**	42.04
AU26	**63.32**	60.50
AU27	82.16	**82.36**
AU28	61.96	**63.80**
AU43	84.09	**84.69**
Average	54.08	**56.02**

B Distribution of Facial Action Units

K-12 Data AU frequency Counts Figure presents the frequency distribution of the 16 facial action units in the classroom dataset. The distribution is highly imbalanced: frequently occurring AUs, particularly AU27 and AU43, contain substantially more positive instances than rare AUs such as AU9, and AU15. Consequently, the model receives considerably less supervision for minority AUs, making their activation patterns more difficult to learn and resulting in lower per-AU F1 scores. This imbalance also explains why overall accuracy can remain high while macro-F1 is substantially lower. Because most AUs are inactive in most frames, a model may achieve high accuracy by correctly predicting the dominant negative class, even when it performs poorly on positive AU instances. Macro-F1 is therefore the primary evaluation metric in this study because it calculates the F1 score independently for each AU and assigns equal importance to both frequent and infrequent AUs. Accordingly, macro-F1 provides a more informative and conservative assessment of model performance under the observed long-tailed AU distribution.

C Action Units to Learning Centred Emotions Mapping

While the proposed model is capable of recognizing an extended set of 16 AUs relevant to K–12 affect analysis, in this preliminary analysis of mapping AUs to learning-centered emotions, we considered only those AUs that have been reported more frequently in association with the corresponding affective states in learning-centered affect literature. This conservative mapping avoids treating infrequently reported AU associations as equally representative and does not assume that the presence of a particular AU, or AU combination, deterministically defines an emotion. The literature-supported AU associations considered in this analysis are summarized in Table 8. A more detailed analysis of AU-to-learning-centered-emotion relationships and mapping strategies across all 16 AUs, including AUs reported more and less frequently in association with these emotions, is presented in the companion study [5].

D Detailed Representation Learning with a Contrastive Objective

This section provides additional implementation details for the AU-specific contrastive representation learning described in the main paper. Unlike conventional image-level contrastive learning, our formulation operates independently on the representation associated with each AU. Given an input image a, the AFG produces one representation $v_i^{(a)}$ for every AU $i \in \{1, \ldots, N\}$. Each representation is passed through a two-layer projection head $g(\cdot)$ and ℓ_2-normalized:

$$z_i^{(a)} = \frac{g(v_i^{(a)})}{\|g(v_i^{(a)})\|_2}, \qquad z_i^{(a)} \in \mathbb{R}^{128}. \tag{13}$$

Contrastive comparisons are performed only between embeddings associated with the same AU identity. Embeddings belonging to different AUs are ignored by the contrastive objective; they serve neither as positives nor as negatives. This design is important because multiple AUs can co-occur and may share facial evidence. Treating representations from different AUs as negatives could therefore introduce inappropriate negative pairs and would conflict with the inter-AU dependencies modeled explicitly by the MEFL and GatedGCN components.

For AU i, let

$$\mathcal{A}_i = \{a \mid y_i^{(a)} = 1\}, \qquad \mathcal{I}_i = \{a \mid y_i^{(a)} = 0\}, \tag{14}$$

denote the active and inactive samples for that AU within a mini-batch. Each active sample $a \in \mathcal{A}_i$ is used as an anchor. The remaining active samples, $\mathcal{A}_i \setminus \{a\}$, form its positive set, whereas the inactive samples $\mathcal{I}_i$ form its negative set. An AU is included in the contrastive objective for a mini-batch only when at least two

Table 8: Literature-supported facial action units associated with learning-centered emotions.

Learning-centered emotion	Best-supported AUs	Reference Papers
Confusion	**AU4 – Brow Lowerer;** **AU7 – Lid Tightener**	[13, 34, 15]
Frustration	**AU1 – Inner Brow Raiser;** **AU2 – Outer Brow Raiser;** **AU4 – Brow Lowerer;** **AU14 – Dimpler**	[13, 15, 21, 34]
Engagement / engaged concentration	**AU5 – Upper Lid Raiser**	[43, 21]
Boredom	**AU43 – Eye Closure**	[13, 15, 34]
Delight	**AU7 – Lid Tightener;** **AU26 – Jaw Drop**	[34]

Note. The table summarizes literature-reported associations for AUs. These associations should not be interpreted as deterministic AU-to-emotion classification rules. Additional AUs reported only infrequently in association with these emotions are not considered in this analysis.

Table 9: Comparison of Emotion-Associated AU True-Positive Recognition on the K–12 Test Set

AU	Description	Emotion	Pos.	OpenGraphAU TP	TP/Pos.	Edu-CL-ME-GraphAU (Ours) TP*	TP/Pos.	Additional TP	TP Increase
AU1	Inner Brow Raiser	Frustration	1,298	149	11.48%	≈534	41.14%	+385	+258.4%
AU2	Outer Brow Raiser	Frustration	3,594	415	11.55%	≈2,280	63.44%	+1,865	+449.4%
AU4	Brow Lowerer	Confusion, Frustration	6,015	3,403	56.58%	≈5,293	88.00%	+1,890	+55.5%
AU5	Upper Lid Raiser	Engagement	4,674	507	10.85%	≈2,105	45.04%	+1,598	+315.2%
AU7	Lid Tightener	Confusion, Delight	1,498	854	57.01%	≈766	51.13%	-88	-10.3%
AU14	Dimpler	Frustration	1,587	1	0.06%	≈448	28.23%	+447	+44,700.0%
AU26	Jaw Drop	Delight	10,006	2,507	25.05%	≈7,136	71.32%	+4,629	+184.6%

Note. TP denotes true-positive AU predictions. TP/Pos. corresponds to recall. Here OpenGraphAU is considered for comparison as it is the pretrained model which performed best on K–12 data. Emotion associations are literature-grounded and do not represent direct emotion-classification. Percentage TP increases may be large when the OpenGraphAU true-positive count is very small.

active samples are available, ensuring that every active anchor has at least one positive example.

Supervised Contrastive Learning. The SCL baseline uses the same AU-specific anchor, positive, and negative construction described above. For an active anchor (a, i), the loss is

$$\mathcal{L}_{a,i}^{\mathrm{SCL}} = -\log \frac{\displaystyle\sum_{m \in \mathcal{A}_i \setminus \{a\}} \exp\left(z_i^{(a)\top} z_i^{(m)} / \tau\right)}{\displaystyle\sum_{m \neq a} \exp\left(z_i^{(a)\top} z_i^{(m)} / \tau\right)}, \tag{15}$$

where the denominator contains all other samples associated with the same AU i, including both active and inactive samples. No embeddings from other AU identities are included.

Balanced Contrastive Learning. Direct application of SCL can be affected by the severe active–inactive imbalance present in AU recognition. In several target AUs, active samples constitute only a small fraction of the observations. Consequently, a summed denominator can be dominated by the substantially larger inactive group.

To reduce this effect, the BCL formulation(BCL-CA) used in the main experiments applies class averaging separately to the active and inactive groups. For an active anchor (a, i), we define

$$S_{a,i}^{+} = \frac{1}{|\mathcal{A}_i| - 1} \sum_{m \in \mathcal{A}_i \setminus \{a\}} \exp\left(z_i^{(a)\top} z_i^{(m)} / \tau\right), \tag{16}$$

and

$$S_{a,i}^{-} = \frac{1}{|\mathcal{I}_i|} \sum_{m \in \mathcal{I}_i} \exp\left(z_i^{(a)\top} z_i^{(m)} / \tau\right). \tag{17}$$

The balanced contrastive loss is then

$$\mathcal{L}^{\mathrm{BCL}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \frac{1}{|\mathcal{A}_i|} \sum_{a \in \mathcal{A}_i} -\log \frac{S_{a,i}^{+}}{S_{a,i}^{+} + S_{a,i}^{-}}, \tag{18}$$

where $\mathcal{M}$ denotes the set of AUs containing at least two active samples in the mini-batch. By averaging the active and inactive groups separately, their contributions are not determined directly by the number of samples in each group.

Thus, SCL and BCL differ only in the aggregation of similarities; the anchor, positive, and negative definitions are unchanged. Both objectives operate independently within each AU identity.

The contrastive projection head is used only during training and is discarded at inference. The same contrastive term is incorporated in both stages of the ME-GraphAU training procedure, with $\gamma = 0.1$ and $\tau = 0.07$.